

MSC2050 Discrete Mathematics, Presentation 4

Dr. Anna Tomskova

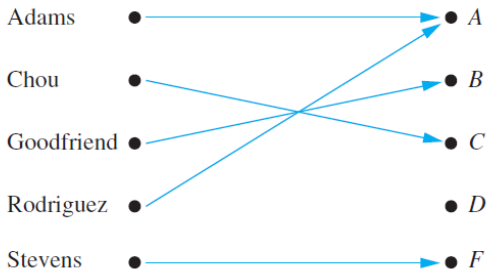


Inha University in Tashkent

Spring 2024
Version 1.0 typed under $\text{\LaTeX}$

Functions

In many instances we assign to each element of a set a particular element of a second set.



Let A and B be nonempty sets.

A **function** f from A to B (denoted as $f : A \rightarrow B$) is an assignment of **exactly one** element of B to each element of A .

We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .

Element a is called **preimage** of b , and b is called an **image** of a .

- Functions are sometimes also called **mappings** or **transformations**.
- Set A is called **domain**.
- Set B is called **codomain**.
- Set $f(A) = \{b \mid \exists a \in A \ f(a) = b\}$ is called **range or image** of the function $f : A \rightarrow B$.
- Image is a subset of codomain, that is $f(A) \subseteq B$.

Examples

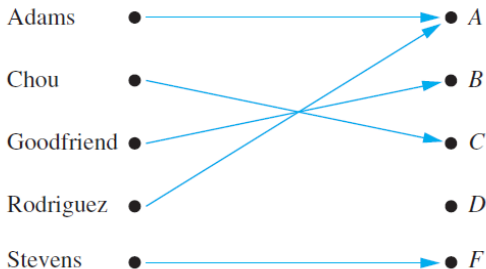


image of "Chou" is "C"

preimage of "F" is "Stevens"

preimage of "A" is {Adams, Rodriguez}

domain is {Adams, Chou, Goodfriend, Rodriguez, Stevens}

codomain is {A, B, C, D, F}

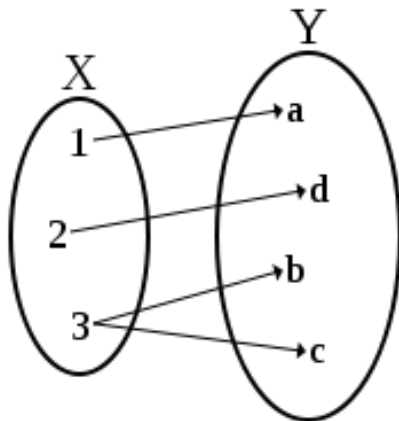
range is {A, B, C, F}

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = x^2$.

Image is $[0, +\infty)$ and is a proper subset of codomain $\mathbb{R}$.

Answer Q1

Examples of not functions



Consider $y = f(x)$, where $x = y^2$ is not a function since $f(1) = 1$ and $f(1) = -1$.

Answer Q2

Let f and g be functions from A to $\mathbb{R}$.

Then $f + g$ and fg are also functions from A to $\mathbb{R}$ defined for all $x \in A$ by

$$(f + g)(x) = f(x) + g(x),$$
$$(fg)(x) = f(x)g(x).$$

For example let f_1 and f_2 be functions on $\mathbb{R}$ (from $\mathbb{R}$ to $\mathbb{R}$) such that

$$f_1(x) = x^2 \text{ and } f_2(x) = x - x^2.$$

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + x - x^2 = x$$

$$(f_1 f_2)(x) = f_1(x)f_2(x) = x^2(x - x^2) = x^3 - x^4$$

Question: Define the function $f_1 + f_2$, where $f_1(x) = \sqrt{x}$ and $f_2(x) = \frac{1}{x}$.

domain of f_1 is $\{x|x \geq 0\}$

domain of f_2 is $\{x|x \neq 0\}$

to define $f_1 + f_2$ both functions must be defined on the same domain

which is $\{x|x \geq 0\} \cap \{x|x \neq 0\} = \{x|x > 0\}$

$$(f_1 + f_2)(x) = \sqrt{x} + \frac{1}{x}$$

Question: Is it possible to define $f_1 f_2$, where $f_1(x) = \sqrt{-x}$ and $f_2(x) = \sqrt{x-1}$?

No

A function f is said to be **one-to-one** if and only if
 $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f .

A function is said to be **injective** if it is one-to-one.

$$\forall a \forall b (f(a) = f(b) \rightarrow a = b)$$

The function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$ and $f(d) = 3$ is one-to-one.

$f : \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(n) = n^2$ is NOT one-to-one, since $f(1) = f(-1) = 1$.
but if restrict the domain $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}$, then it is one-to-one.

$f(x) = x + 1$ of $\mathbb{R}$ is one-to-one. If $f(a) = f(b)$, then $a + 1 = b + 1$ or $a = b$.

A function f is

increasing if $\forall x \forall y (x < y \rightarrow f(x) \leq f(y))$

strictly increasing if $\forall x \forall y (x < y \rightarrow f(x) < f(y))$

decreasing if $\forall x \forall y (x < y \rightarrow f(x) \geq f(y))$

strictly decreasing if $\forall x \forall y (x < y \rightarrow f(x) > f(y))$

Strictly decreasing and strictly increasing functions are one-to-one. Answer Q3

A function f from A to B is called **onto**, if and only if
for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.

A function f is called **surjective** if it is onto.

$$\forall y \exists x (f(x) = y)$$

The function f from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ with $f(a) = 3$, $f(b) = 2$, $f(c) = 1$ and $f(d) = 3$ is onto.

$f : \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(n) = n^2$ is NOT onto, since there is no $x \in \mathbb{Z}$ with $f(x) = -1$.
but if change the codomain $f : \mathbb{Z} \rightarrow \mathbb{Z}^+$, then it is onto.

$f(x) = x + 1$ of $\mathbb{R}$ is onto. Since for any $y \in \mathbb{R}$ there is $x = y - 1$ such that
 $f(x) = (y - 1) + 1 = y$.

Answer Q4

Summary

Suppose that $f : A \rightarrow B$.

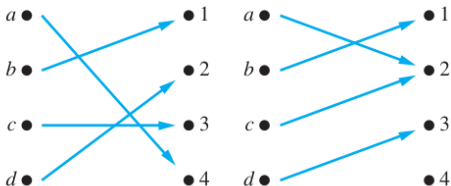
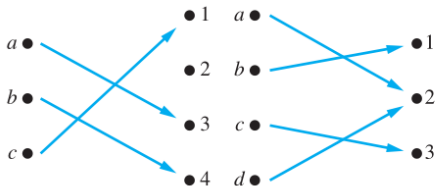
To show that f is **injective** show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

To show that f is not **injective** find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

To show that f is **surjective** consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

To show that f is not **surjective** find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

A function that is injective and surjective at the same time is called **bijective** function, or *bijection*.



One-to-one, not onto onto, not one-to-one bijection not one-to-one, not onto

Inverse functions

Let f be a **bijection** from the set A to the set B .

The **inverse function** of f is the function that assigns to an element $b \in B$ the unique element $a \in A$ such that $f(a) = b$.

The inverse function of f is denoted by f^{-1} (not the same as $\frac{1}{f}$)

Hence, $f^{-1}(b) = a$ when $f(a) = b$.

Bijection f is called **invertible**, since it has an inverse function.

f from $\{a, b, c\}$ to $\{1, 2, 3\}$ with $f(a) = 3$, $f(b) = 2$, $f(c) = 1$ is invertible.

$f(x) = x + 1$ of $\mathbb{R}$ is invertible. The inverse is $f^{-1}(y) = y - 1$.

$f : \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(n) = n^2$ is NOT invertible.

but $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ is invertible.

Answer Q5

Composition

Let $g : A \rightarrow B$ and $f : B \rightarrow C$.

The **composition** of f and g is the function $f \circ g : A \rightarrow C$ defined by

$$(f \circ g)(a) = f(g(a))$$

Let $f(x) = 2x + 3$ and $g(x) = x^2$.

$$(f \circ g)(x) = f(g(x)) = f(x^2) = 2x^2 + 3$$

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = (2x + 3)^2 = 4x^2 + 12x + 9$$

$$f \circ g \neq g \circ f.$$

Let $f : A \rightarrow B$ be invertible.

For $a \in A$ with $f(a) = b$ we have

$$(f^{-1} \circ f)(a) = f^{-1}(f(a)) = f^{-1}(b) = a$$

$$(f \circ f^{-1})(b) = f(f^{-1}(b)) = f(a) = b$$

$$f^{-1} \circ f = i_A \quad f \circ f^{-1} = i_B$$

Answer Q6

The graph of functions

Let $f : A \rightarrow B$.

The **graph** of f is the set

$$\{(a, b) \mid a \in A, f(a) = b\}.$$

The graph of the function $f(n) = 2n + 1$ on the set of integers is at Figure 1

The graph of the function $f(n) = n^2$ on the set of integers is at Figure 2

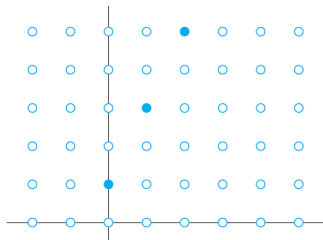


Figure 1

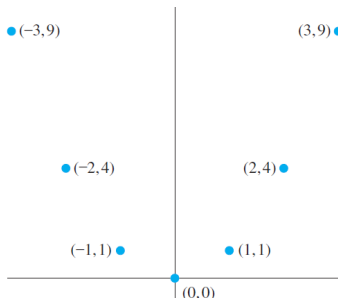


Figure 2

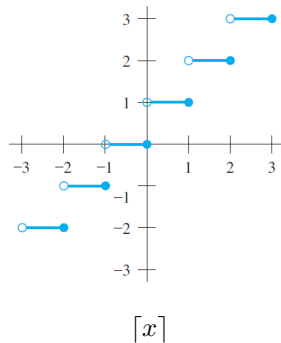
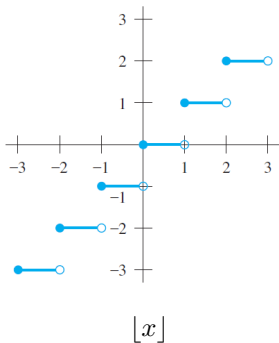
Floor and ceiling functions

The **floor** function assigns to the real number x the largest integer that is less than or equal to x .

The floor function is denoted by $\lfloor x \rfloor$

The **ceiling** function assigns to the real number x the smallest integer that is less than or equal to x .

The ceiling function is denoted by $\lceil x \rceil$



Answer Q7

Homework:

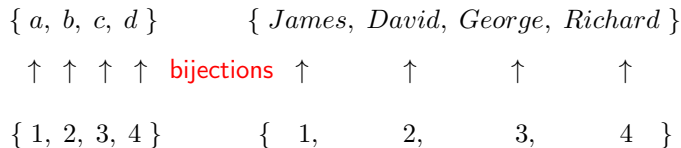
- Solve all *odd* exercises starting at p. 152.

Cardinality of sets

(paragraph 2.5)

The same cardinalities

Question: When two **finite** sets A and B have the same cardinality?



Two (finite or infinite) sets A and B are said to have **the same cardinality** if there is a bijection between them. In this case we write $|A| = |B|$.

Construct a bijection

$$f : \{a, b, c, d\} \rightarrow \{James, David, George, Richard\}$$

$$f(a) = James, f(b) = David, f(c) = George, f(d) = Richard$$

or

$$f(a) = George, f(b) = James, f(c) = Richard, f(d) = David$$

Answer Q8

Different cardinalities

It is impossible to find bijection between sets with different cardinalities.

In this case we write $|A| \leq |B|$. There is a one-to-one function from A to B .

If there is a one-to-one function from A to B , we write $|A| \leq |B|$.
When $|A| \leq |B|$ and A and B have different cardinality, we write $|A| < |B|$.

However, it is not easy to show that two infinite sets have different cardinality. For example, we can construct a one-to-one function from $\mathbb{N}$ to $\mathbb{Z}$, hence $|\mathbb{N}| \leq |\mathbb{Z}|$. Similarly, $|\mathbb{N}| \leq |\mathbb{R}|$.
We will show that $|\mathbb{N}| = |\mathbb{Z}|$ but $|\mathbb{N}| < |\mathbb{R}|$

Answer Q9

Countable sets

We will now split all infinite sets into **two** groups, those with the same cardinality as the set of positive integers and those with a different cardinality.

A set that is either finite or has the same cardinality as the set of positive integers is called **countable**.

A set that is not countable is called **uncountable**.

When an infinite set S is countable, we write $|S| = \aleph_0$ (where $\aleph$ is **aleph**, the first letter of the Hebrew alphabet).

Question: Show that the set of odd positive integers is a countable set.

Solution: Consider the function $f(n) = 2n - 1$ from $\mathbb{N}$ to the set of odd positive integers.

We will show that f is a bijection.

Let $f(n) = f(m)$. Then $2n - 1 = 2m - 1$, so $n = m$. So f is one-to-one.

Suppose that t is an odd positive integer. Then $t = 2n - 1$ for some $n \in \mathbb{N}$.

Hence, $t = f(n)$.

HILBERT'S GRAND HOTEL

The famous mathematician David Hilbert invented the notion of the Grand Hotel, which has a **countably infinite number of rooms**, each occupied by a guest.

When a new guest arrives at a hotel with a finite number of rooms, and all rooms are occupied, this guest cannot be accommodated without evicting a current guest. However, we can always accommodate a new guest at the Grand Hotel, even when all rooms are already occupied

Because the rooms of the Grand Hotel are countable, we can list them as **Room 1, Room 2, Room 3, and so on**. When a new guest arrives, we move the guest in Room 1 to Room 2, the guest in Room 2 to Room 3, and in general, the guest in Room n to Room $n + 1$, for all positive integers n . This frees up Room 1, which we assign to the new guest, and all the current guests still have rooms.

Examples of countable sets

- The set of positive integers $\mathbb{Z}^+$, with bijection $f(n) = n$.
- The set of natural numbers $\mathbb{N}$, with bijection $f(n) = n - 1$
- The set of all integers $\mathbb{Z}$, with bijection $f(n) = \frac{n}{2}$ when n is even and $f(n) = \frac{-(n-1)}{2}$ when n is odd.
- The set of positive rational numbers, with bijection

Answer Q10

Countable union of countable sets

Theorem

If the sets $S_1, S_2, S_3, \dots$ are countable, then the countable union

$$\bigcup_{i=1}^{\infty} S_i$$

is countable.

Example from midterm

- (8 points) Let M_n be the set of $n \times n$ matrices with entries that are either 0 or 1.
- (a) (2 point) Give two examples of matrices from the set M_2 .
- (b) (2 points) Is the set M_2 finite or infinite? Justify.
- (c) (4 points) Let M be a set of all square matrices (of all sizes) with entries, that are either zero or one. Is the set M finite, countably infinite or uncountable? Justify.

Uncountable sets

Theorem

The interval $(0, 1)$ is an uncountable set

Proof: (Cantor diagonalization argument)

Assume that $(0, 1)$ is countable. It means that we can write them all as

$$r_1 = 0.d_{11}d_{12}d_{13}d_{14} \dots$$

$$r_2 = 0.d_{21}d_{22}d_{23}d_{24} \dots$$

$$r_3 = 0.d_{31}d_{32}d_{33}d_{34} \dots$$

$$r_4 = 0.d_{41}d_{42}d_{43}d_{44} \dots$$

$\vdots$

where $d_{ij} \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Let $r = 0.d_1d_2d_3d_4 \dots$, where

$$d_i = \begin{cases} 4, & d_{ii} \neq 4 \\ 5 & d_{ii} = 4 \end{cases}$$

r is in $(0, 1)$ but $r \neq r_i$ for all i . Hence, $(0, 1)$ is not countable.

Examples of uncountable sets

Show that $|(0, \infty)| = |(0, 1)|$

Consider the function $f : (0, \infty) \rightarrow (0, 1)$ given by

$$f(x) = \frac{x}{x+1}.$$

Show that $|\mathbb{R}| = |(0, 1)|$

Show that $|\mathbb{R}| = |(0, \infty)|$.

Consider the function $f : \mathbb{R} \rightarrow (0, \infty)$ given by

$$f(x) = 2^x.$$

SCHRÖDER-BERNSTEIN THEOREM

If A and B are sets with $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.

In other words, if there are one-to-one functions f from A to B and g from B to A , then $|A| = |B|$.

Show that $|(0, 1]| = |(0, 1)|$

$f(x) = x$ is a one-to-one function from $(0, 1)$ to $(0, 1]$.

The function $g(x) = x/2$ is one-to-one from $(0, 1]$ to $(0, 1/2] \subset (0, 1)$.

By the S-B theorem, $|(0, 1)| = |(0, 1]|$.

Answer Q11

The Set of all Subsets of Natural Numbers is Uncountable

Theorem 1: The set $\mathcal{P}(\mathbb{N})$ of all subsets of $\mathbb{N}$ is uncountable.

In the proof below, we use the famous diagonalization argument to show that the set of all subsets of $\mathbb{N}$ is uncountable.

- **Proof:** Suppose that $\mathcal{P}(\mathbb{N})$ is countable.
- We can denote each set $A \subseteq \mathbb{N}$ by a decimal number of the form:

$$0.x_0x_1 \dots x_n \dots \quad (1)$$

- where each $x_n \in \{0, 1\}$ and such that $x_n = 0$ if $n \in A$ and $x_n = 1$ if $n \notin A$. For example, the set $\{1, 3, 4, 5, 6\}$ has decimal representation 0.1011110.... Now since we assume that $\mathcal{P}(\mathbb{N})$ is countable, there exists a bijective function $f : \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$ defined for all $n \in \mathbb{N}$ by $f(n) = A_n$. We list the sets $\{A_1, A_2, \dots, A_n, \dots\}$ and their decimal representations as follows:

$$\begin{array}{l} A_1, \quad 0.x_{1,1}x_{1,2} \dots x_{1,n} \dots \\ A_2, \quad 0.x_{2,1}x_{2,2} \dots x_{2,n} \dots \\ \vdots \\ A_n, \quad 0.x_{n,1}x_{n,2} \dots x_{n,n} \dots \\ \vdots \end{array} \quad (2)$$

- Now construct a decimal number $x = 0.x_1x_2 \dots x_n \dots$ as follows. We let $x_1 = 0$ if $x_{1,1} = 1$ and we let $x_1 = 1$ if $x_{1,1} = 0$. We let $x_2 = 0$ if $x_{2,2} = 1$ and we let $x_2 = 1$ if $x_{2,2} = 0$. In general, we let $x_n = 0$ if $x_{n,n} = 1$ and we let $x_n = 1$ if $x_{n,n} = 0$. Then X differs from all of the decimal numbers in the list above in at least one decimal spot. So the decimal x corresponds to a subset X of $\mathbb{N}$ that is not on the list above. This is a contradiction since f is assumed to be a bijection.
- Therefore the set $\mathcal{P}(\mathbb{N})$ is uncountable. ■

Uncomputable functions

We say that a function is **computable** if there is a computer program in some programming language that finds the values of this function.
If a function is not computable we say it is **uncomputable**.

To show that there are uncomputable functions, we need to establish two results:

- 1 The set of all computer programs in any particular programming language is countable (a computer program is a string of characters from a finite alphabet)
- 2 the set of functions from the set of positive integers to itself is uncountable (construct a one-to-one correspondence between the set $(0, 1)$ and a subset of these functions. Do this by associating to the real number $0.d_1d_2 \dots d_n \dots$ the function $f : \mathbb{Z}^+ \rightarrow \{0, 1, \dots, 9\}$ with $f(n) = d_n$.)

The continuum hypothesis

We know that $\mathbb{N}$ is countable that is

$$|\mathbb{N}| = \aleph_0$$

The power set $\mathcal{P}(\mathbb{N})$ is uncountable, that is

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = 2^{\aleph_0} = c$$

The continuum hypothesis

There is **no** set A such that

$$\aleph_0 < |A| < c$$

The continuum hypothesis was advanced by Georg Cantor in 1878, and published as the first of Hilbert's 23 problems presented in 1900.

Homework:

- paragraphs 2.4 (sequences) and 2.6 (matrices) are left for independent study
- Solve *odd* exercises starting at p. 176, #1-#25 and #33, #35.